

DAY-4

Components specific troubleshooting practical

1. Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.

RAM Reseating Procedure

Objective: To safely remove and re-insert a RAM module and check that it is properly seated.

1. *Shut down the computer completely and disconnect the power cable.*
2. *Press the power button for a few seconds after unplugging to discharge any remaining power.*
3. *Open the computer case by removing the side panel. For a laptop, remove the bottom cover if the RAM is accessible.*

4. *Locate the RAM slots on the motherboard. They are usually long slots located near the CPU socket.*
5. *Ground yourself by touching the metal chassis to reduce the risk of static electricity.*
6. *Release the RAM module by pressing the locking clips at both ends of the RAM slot.*
7. *Carefully remove the RAM module by holding it only by its edges. Avoid touching the gold contacts.*
8. *Inspect the RAM and slot for dust, damage, or bent parts.*
9. *Align the notch on the RAM module with the key in the RAM slot.*
10. *Re-insert the RAM evenly into the slot and press down firmly until the locking clips click into place.*
11. *Check the RAM to make sure both locking clips are properly secured.*
12. *Close the computer case, reconnect the power cable, and turn on the computer.*
13. *Check for successful POST/boot and verify that the system recognizes the installed RAM in the BIOS/UEFI or operating system.*

Issues Observed

- *No physical damage was observed on the RAM module or slot.*
- *The RAM was removed and re-inserted successfully.*
- *The locking clips clicked into place after reseating.*
- *The computer booted normally after the procedure.*
- *The installed RAM was detected correctly by the system.*

***Safety note:** Always switch off and unplug the computer before reseating RAM, and handle the module by its edges to avoid damaging it with static electricity or fingerprints.*

2. Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix.

***Hint:** Think about display issues, driver checks, and physical connections.*

1. Identify and document symptoms

Note exactly what's happening: black screen, no display signal, artifacts (colored dots/lines/flickering), crashes during gaming but not desktop use, driver crashes, unusual fan noise, or burning smell. Note WHEN it happens — at boot, under load, or randomly — since this narrows down whether it's a GPU, PSU, or motherboard issue.

2. Check physical connections

Power down and unplug the PC. Reseat the GPU firmly in its PCIe slot until it clicks. Check that all PCIe power connectors (6-pin/8-pin) from the PSU are fully seated into the card. Verify the monitor cable is connected directly to the GPU's output, not the motherboard's onboard video port.

3. Inspect for visible damage or overheating

Look for bent pins, scorch marks, bulging capacitors, or dust buildup clogging the fans/heatsink. Feel if the card gets excessively hot during light use. Clean out dust with compressed air, as clogged fans are a common cause of overheating and crashes.

4. Test with minimal hardware and onboard graphics

If your CPU has integrated graphics, remove the GPU entirely and connect the monitor to the motherboard's video output. If the PC displays fine without the GPU, this strongly points to the GPU itself. If there's still no display, the issue likely lies elsewhere (PSU, motherboard, RAM).

5. Update or roll back GPU drivers

Boot into Safe Mode and use a tool like DDU (Display Driver Uninstaller) to fully remove existing GPU drivers, then install the latest stable driver from the manufacturer's website (NVIDIA/AMD/Intel). If crashes started after a recent driver update, try rolling back to a previous stable version instead.

6. Monitor temperatures and run a stress test

Use software like MSI Afterburner or HWiNFO to monitor GPU temperature, clock speeds, and fan RPM in real time. Run a stress test (e.g., FurMark or a demanding game) to see if crashes, artifacts, or thermal throttling occur under load — this helps confirm whether heat or instability is the root cause.

7. Test the GPU in another system (if possible)

If you have access to a second PC, install the suspect GPU there. If the same symptoms appear, the GPU is very likely faulty. If it works fine

elsewhere, the problem is more likely with your original PC's PSU, motherboard, or power delivery.

8. Confirm the fix

Once you've made a change (reseating, driver update, cleaning, or swapping hardware), run the same stress test or game session that originally caused issues for at least 30-60 minutes. If temperatures stay stable, no artifacts appear, and no crashes occur, the issue is resolved. If problems persist after ruling out drivers, connections, and other hardware, the GPU likely needs RMA or replacement.

3. Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.

Short Storage Test

- 1. Use a secondary test SATA/NVMe drive—not the boot drive.*
- 2. Shut down, unplug power, and disconnect the drive.*
- 3. Boot the PC and check BIOS:*
 - Drive should be missing.*
 - If it was the boot drive, you may see “No bootable device” or “Operating system not found.”*
- 4. Check Windows Disk Management:*

- *The drive should be absent.*
 - 5. *Power off and reconnect/reseat the drive.*
 - 6. *Boot again and verify:*
 - *Drive model and capacity return in BIOS.*
 - *The drive appears in Disk Management/File Explorer.*
 - *Files open normally.*
4. Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.

RAM diagnostic steps

1. Check Event Viewer

- *Open Event Viewer → Windows Logs → System.*
- *Note errors around the restart, especially memory-diagnostic or WHEA hardware errors.*

2. Disable memory overclocking

- *Enter BIOS/UEFI and disable XMP/EXPO.*
- *Load default RAM timings and voltage.*

3. Reseat the RAM

- *Shut down, unplug power, and discharge the PC.*
- *Remove and reinstall each module firmly.*

4. Test one module at a time

- *Boot with only one RAM stick installed.*
- *Repeat with the other stick and try the recommended slots.*

5. Run a memory test

- *Use Windows Memory Diagnostic or bootable MemTest86.*
- *MemTest86 recommends testing modules individually to identify a failing stick or slot.*

6. Confirm the result

- *Run several test passes and use the PC or game that previously caused restarts.*
- *If errors occur with one module in multiple slots, replace that RAM module.*
- *If all RAM tests pass, investigate the GPU, temperatures, PSU, and motherboard next.*

A restart occurring only during gaming would make the GPU, GPU driver, temperatures, and PSU more urgent checks than RAM.